

Final Report Template

Hotel Booking Demand Clustering Study

[Team name] - [Student 1], [Student 2], [Student 3]

Group number

Unsupervised Learning - 2025/2026 |

Repository tag: [Submission] |

Date: [DD Month 2026]

Use this file as a concise paper structure. Replace bracketed text with your own writing and remove guidance notes before submission. Target length: 10 pages, single-column, excluding references and appendix.

Abstract

One paragraph, about 150-200 words. Summarise the question, data, method, evaluation evidence, main finding, and key limitation.

[Abstract text.]

Keywords: [hotel booking demand; clustering; segmentation; mixed-type data; reproducibility]

1. Introduction

Motivate the hotel booking segmentation problem and state the concrete question answered by the report.

- Context: why grouping booking records may be useful.
- Segmentation question: one clear question that can be studied without class labels.
- Clustering rationale: what similarity between bookings means in this study.
- Contribution: the main pipeline, clustering methods, evaluation, and interpretation.

2. Dataset and Preprocessing

2.1 Dataset documentation

Give enough information for traceability, without reproducing the full data

Item	Report here
Dataset and version	[Hotel Booking Demand; course release/version used]
Unit of analysis	[One row = one booking record]
Size and time span	[Rows, columns, date range if applicable]
Source and terms	[Source, citation, license/usage restrictions]
Checksum/hash	[Hash of the raw file used]
Raw-data handling	[State that raw data are not committed; give preparation instructions]

2.2 Data quality and feature governance

Briefly report missingness, outliers, exclusions, and the final feature set.

- Summarise numerical and categorical missingness and outlier checks.
- Define the segmentation/index time and exclude post-event or outcome variables from clustering inputs.
- Document how high-cardinality or ID-like fields are handled.
- State the final representation and the distance/similarity implied by it.

3. Clustering Methods

Describe the models at a level that allows reproduction.

Keep detailed parameter tables in the repository or appendix if needed.

All methods should be compared using the same data representation unless a different representation is explicitly justified.

4. Experimental Protocol

Explain how the comparison was made fair and reproducible.

- Predefine the parameter ranges and justify them briefly.
- Use repeated seeds or resampling where relevant, and report variability.
- Report Silhouette plus one/two additional internal validity index.
- Include one justification suited to the alternative clustering family.
- Include at least one controlled sensitivity check, such as a preprocessing or representation variant.
- Record experiments and ensure that figures and tables can be reproduced from the repository.

Experiment block	Purpose	Main output
Baseline models	...	[Model comparison table and cluster profiles]
Alternative family	[Conceptually different clustering approach]	[Diagnostic plot and comparison with baseline]
Robustness checks	[Seeds/resampling and one controlled variant]	[Stability and sensitivity summary]

5. Results and Cluster Interpretation

5.1 Data preparation outcomes

Report the final analytic dataset: rows, selected features, encoded dimensions, and main exclusions.

5.2 Model comparison

Present one compact comparison table. Explain the selected model;

Method	Selected setting	Validity evidence	Stability evidence	Main observation
[Baseline]	[...]	[...]	[...]	[...]
[iK-means]	[...]	[...]	[...]	[...]
[Alternative]	[...]	[...]	[...]	[...]

Possible suggestions

Figure 1. [Pipeline or model figure.]

Figure 2. [Main clustering result or cluster-profile visual summary.]

5.3 Cluster profiles

Translate clusters into booking-profile narrative using descriptive statistics and visual summaries.

Outcome or post-event fields may be used here only for post-hoc interpretation.

Cluster	Size	Main characteristics	Post-hoc interpretation	Profile label
C0	[n, %]	[...]	[...]	[...]
C1	[n, %]	[...]	[...]	[...]

C2	[n, %]	[...]	[...]	[...]
C3+	[n, %]	[...]	[...]	[...]

6. Discussion

Answer the research questions directly and connect the empirical evidence to the original problem.

- Which clustering result best answers the segmentation question?
- How did preprocessing and similarity choices influence the results?
- Are the clusters coherent, stable, and interpretable enough to be useful?
- What decision-relevant hypotheses or recommendations are supported by the evidence?
- What should not be concluded from this unsupervised analysis?

[Write this section here.]

7. Limitations, Ethics, and Reproducibility

Keep this section concise but explicit.

- Limitations: no ground-truth labels ...
- Reproducibility: report the repository tag, experiment log, environment file, run_all entry point, and reproducible figures/tables.
- Ethics and governance: follow dataset terms, do not commit the raw data, ...
- Academic integrity: include any declarations required by the course rules.

[Write this section here.]

8. Conclusion

One short paragraph: restate the question, the main result, robustness judgement, and next step.

References

Use one consistent citation style. Include the dataset paper, dataset/source documentation, and any methods or libraries cited in the report.

António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41-49.
<https://doi.org/10.1016/j.dib.2018.11.126>

[Additional references.]